DATA READINESS SYSTEM (DRS)

SYSTEM REQUIREMENTS SPECIFICATION (SRS)

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1. INTRODUCTION

1.1 Purpose

This System Requirements Specification (SRS) defines the complete functional and non-functional requirements for the Data Readiness System (DRS). The purpose is to ensure the system is implementable, testable, and verifiable.

1.2 Scope

The DRS system shall support:

- Physical archive digitization (factory workflow)
- Digital ingestion (connectors)
- Immutable evidence storage
- Document intelligence (OCR + extraction)
- Trust scoring and validation workflows
- Canonical Common Data Model mapping
- Entity resolution and golden record creation
- Knowledge graph generation
- Dataset publishing and versioning
- Governance and compliance controls
- Multi-tenant, on-prem, and air-gapped deployments

1.3 Intended Audience

This SRS is intended for:

- Systems engineers
- Software engineers
- AI/ML engineers
- QA and testing teams
- Compliance officers
- DevOps engineers
- Project stakeholders

1.4 Definitions and Acronyms

(Refer to DDD definitions.)

2. SYSTEM OVERVIEW

2.1 System Mission

The DRS shall convert raw paper and digital evidence into structured, validated, traceable datasets for AI and decision-making systems.

2.2 System Boundary

The system includes:

- ingestion mechanisms
- evidence storage
- processing pipelines
- validation workflows
- publishing mechanisms

The system excludes:

- downstream AI model training pipelines (DRS provides datasets only)
- enterprise system modification unless explicitly integrated

3. REQUIREMENT CATEGORIES

Requirements are classified as:

- Functional Requirements (FR)
- Non-Functional Requirements (NFR)
- Security Requirements (SEC)
- Integration Requirements (INT)
- Data Requirements (DATA)
- Operations Requirements (OPS)
- Verification Requirements (VV)

4. FUNCTIONAL REQUIREMENTS (FR)

4.1 Physical Operations and Custody Requirements

FR-01: Batch Registration

The system shall allow an operator to create and register a new digitization batch with a unique Batch ID.

FR-02: Folder Registration

The system shall allow an operator to register folders under a batch with unique Folder IDs.

FR-03: Document Registration

The system shall allow registration of individual documents under folders.

FR-04: QR Code Generation

The system shall generate QR codes for batches, folders, and documents.

FR-05: Physical Location Tracking

The system shall record physical storage location metadata including box, shelf, room, and building identifiers.

FR-06: Scan Completion Reconciliation

The system shall support reconciliation of expected vs scanned document/page counts for each batch.

FR-07: Custody Chain Logging

The system shall generate a custody audit record for every batch handling event.

4.2 Ingestion and Evidence Requirements

FR-08: Evidence File Upload

The system shall support uploading scanned files (PDF/images) into the Evidence Layer.

FR-09: Digital Source Ingestion

The system shall support ingestion of digital files from external sources such as SharePoint, cloud drives, email, and database exports.

FR-10: Evidence Hash Generation

The system shall generate a cryptographic SHA-256 hash for every ingested evidence file.

FR-11: Immutable Evidence Locking

The system shall enforce immutability for all evidence objects after ingestion.

FR-12: Evidence Metadata Storage

The system shall store provenance metadata for each evidence object including source type, timestamp, uploader identity, and source reference.

4.3 Document Intelligence Requirements

FR-13: Document Type Classification

The system shall classify documents into predefined document types with confidence scores.

FR-14: OCR Extraction

The system shall perform OCR extraction on scanned documents and images.

FR-15: Layout Detection

The system shall support layout detection for text blocks, tables, stamps, and signatures.

FR-16: Table Extraction

The system shall extract tabular data from documents into structured table objects.

FR-17: Named Entity Extraction

The system shall extract named entities including persons, organizations, dates, IDs, and financial amounts.

FR-18: Evidence Pointer Generation

The system shall generate evidence pointers linking extracted fields to page/coordinate regions.

4.4 Trust Scoring and Validation Requirements

FR-19: Confidence Scoring

The system shall generate confidence scores for each extracted field.

FR-20: Trust Scoring

The system shall compute an overall trust score per record object using configurable weights.

FR-21: Validation Task Creation

The system shall generate validation tasks for records with trust score below a configurable threshold.

FR-22: Human Validation Workflow

The system shall provide a workflow interface for validators to approve, reject, or request rescan for extracted fields.

FR-23: Multi-Approver Validation

The system shall support multi-stage approval workflows for sensitive datasets.

FR-24: Validation History Tracking

The system shall store validation history for each record object and each validated field.

4.5 Canonical Data Model Requirements

FR-25: CDM Definition

The system shall implement a canonical Common Data Model (CDM) including at minimum Person, Organization, Document, Transaction, and Asset entities.

FR-26: CDM Mapping

The system shall map validated extracted records into CDM entity objects.

FR-27: Schema Versioning

The system shall support versioned schemas for the CDM.

FR-28: Tenant Schema Extension

The system shall allow tenant-specific schema extensions without breaking core CDM compatibility.

4.6 Knowledge Synchronization and Entity Resolution Requirements

FR-29: Entity Matching

The system shall identify potential duplicate entities using exact match and fuzzy match methods.

FR-30: Entity Resolution Review Queue

The system shall support a review queue for ambiguous entity matches.

FR-31: Golden Record Creation

The system shall create golden records by merging multiple entity candidates based on merge rules.

FR-32: Merge Rule Enforcement

The system shall enforce configurable merge rules for golden record field selection.

FR-33: Knowledge Graph Generation

The system shall generate a knowledge graph representing entities and their relationships.

FR-34: Knowledge Graph Update

The system shall update the knowledge graph whenever a validated CDM entity is added or modified.

FR-35: Synchronization with External Systems

The system shall support optional synchronization of golden records back into enterprise systems when enabled.

4.7 Dataset Publishing Requirements

FR-36: Dataset Generation

The system shall generate datasets from CDM entities and knowledge graph records.

FR-37: Dataset Format Export

The system shall export datasets in CSV, JSONL, and Parquet formats.

FR-38: Dataset Versioning

The system shall version datasets using semantic versioning.

FR-39: Dataset Card Generation

The system shall generate dataset cards containing schema, lineage, quality metrics, and validation signoff.

FR-40: Dataset Download Access

The system shall allow authorized users to download published datasets.

FR-41: Dataset Regeneration

The system shall support regeneration of dataset versions using stored evidence and processing configurations.

4.8 Governance and Compliance Requirements

FR-42: RBAC Enforcement

The system shall enforce role-based access control for all system functions.

FR-43: Audit Log Generation

The system shall generate audit logs for all significant system actions including ingestion, validation, publishing, and access.

FR-44: Retention Policy Management

The system shall allow configuration of retention policies for evidence and datasets.

FR-45: PII Detection Support

The system shall support classification of fields as PII for governance enforcement.

4.9 Platform Operations Requirements

FR-46: Multi-Tenant Isolation

The system shall support multiple tenants with strict data isolation.

FR-47: Monitoring and Observability

The system shall provide system monitoring metrics for ingestion throughput, processing latency, storage usage, and queue backlog.

FR-48: Backup and Recovery

The system shall support scheduled backups and restoration for all system databases and evidence metadata.

FR-49: Model Version Control

The system shall store version identifiers for all AI models used in extraction workflows.

FR-50: Workflow Notification

The system shall support notifications to users when validation tasks are assigned or completed.

5. DATA REQUIREMENTS (DATA)

DATA-01: Evidence Storage Integrity

The system shall store all evidence objects in immutable object storage with hash verification.

DATA-02: Evidence Metadata Persistence

The system shall store metadata for each evidence object in a relational database.

DATA-03: Lineage Traceability

The system shall support full lineage traceability from dataset value to raw evidence source.

DATA-04: Evidence Pointer Format

The system shall store evidence pointers in a standardized format referencing page and bounding box coordinates.

DATA-05: Record Object Storage

The system shall store Standard Record Objects (SROs) containing extracted candidate fields and confidence values.

DATA-06: Validation Data Storage

The system shall store validation results including validator identity and timestamp.

DATA-07: Knowledge Graph Storage

The system shall store entity relationships in a graph database.

DATA-08: Dataset Storage

The system shall store published datasets in a versioned storage location.

DATA-09: Dataset Card Storage

The system shall store dataset cards as structured metadata objects.

DATA-10: Audit Log Storage

The system shall store audit logs in a tamper-evident storage system.

6. INTEGRATION REQUIREMENTS (INT)

INT-01: Scanner Interface Support

The system shall support ingestion of scanner output formats including PDF, TIFF, PNG, and JPEG.

INT-02: Object Storage Compatibility

The system shall support S3-compatible object storage interfaces.

INT-03: Identity Provider Integration

The system shall integrate with external identity providers using OAuth2 / OpenID Connect.

INT-04: External API Connector Framework

The system shall provide a connector framework for integrating with ERP, CRM, and document management systems.

INT-05: Connector Scheduling

The system shall support scheduled ingestion jobs for connectors.

INT-06: Connector Error Handling

The system shall log connector failures and retry ingestion jobs.

INT-07: Dataset Export Integration

The system shall provide export endpoints usable by external AI pipelines.

INT-08: Logging Integration

The system shall integrate with centralized logging tools such as ELK or OpenSearch.

7. SECURITY REQUIREMENTS (SEC)

SEC-01: Authentication

The system shall require authenticated access for all users.

SEC-02: Authorization

The system shall enforce RBAC authorization for all API endpoints and user actions.

SEC-03: Encryption at Rest

The system shall encrypt all stored evidence and database records at rest using AES-256 or equivalent.

SEC-04: Encryption in Transit

The system shall enforce TLS 1.2+ encryption for all network communication.

SEC-05: Key Management

The system shall support integration with KMS or Vault for secure key management.

SEC-06: Multi-Factor Authentication

The system shall support MFA for privileged accounts.

SEC-07: Audit Logging

The system shall log all security-relevant actions including login attempts, role changes, and dataset access.

SEC-08: Access Logging for Evidence

The system shall log every access attempt to evidence objects.

SEC-09: Tenant Data Isolation

The system shall ensure tenant data cannot be accessed by other tenants.

SEC-10: Admin Privilege Controls

The system shall enforce least-privilege rules for administrative actions.

SEC-11: Data Export Controls

The system shall restrict dataset export based on role and dataset classification.

SEC-12: Secure Deletion Policy

The system shall support secure deletion mechanisms for tenant data when required by policy.

8. NON-FUNCTIONAL REQUIREMENTS (NFR)

NFR-01: Availability

The system shall provide a minimum uptime of 99.5% in SaaS deployments.

NFR-02: Scalability

The system shall support horizontal scaling for ingestion and document processing services.

NFR-03: Performance - Ingestion

The system shall ingest at least 1,000 documents per hour per tenant under nominal load (configurable target).

NFR-04: Performance - OCR Processing

The system shall process OCR for a scanned page within 10 seconds under normal operating conditions.

NFR-05: Latency - Dataset Export

The system shall generate a dataset export within 10 minutes for datasets under 1 million records.

NFR-06: Data Consistency

The system shall ensure consistent dataset outputs for the same evidence and configuration inputs.

NFR-07: Maintainability

The system shall be modular such that each subsystem can be updated independently.

NFR-08: Extensibility

The system shall support adding new connector types without modification of core ingestion logic.

NFR-09: Reliability

The system shall provide automatic retry mechanisms for failed pipeline jobs.

NFR-10: Fault Tolerance

The system shall prevent partial failures from corrupting stored evidence or datasets.

NFR-11: Portability

The system shall support deployment in cloud, on-prem, and air-gapped environments.

NFR-12: Auditability

The system shall ensure every dataset value can be traced back to source evidence.

NFR-13: Usability

The system shall provide user interfaces for operators, validators, compliance officers, and AI engineers.

NFR-14: Interoperability

The system shall support standard data export formats usable by downstream systems.

NFR-15: Logging

The system shall log all pipeline events and errors for debugging and audit.

9. OPERATIONAL REQUIREMENTS (OPS)

OPS-01: Deployment Automation

The system shall support automated deployment using Docker and Kubernetes manifests.

OPS-02: Backup Scheduling

The system shall support configurable automated backup schedules.

OPS-03: Disaster Recovery

The system shall provide restoration procedures for evidence metadata, datasets, and audit logs.

OPS-04: Monitoring Dashboards

The system shall provide dashboards for monitoring system health and throughput.

OPS-05: Air-Gapped Operation

The system shall operate without external internet dependencies in air-gapped deployments.

OPS-06: Offline Model Management

The system shall support manual uploading and version control of AI models in air-gapped deployments.

OPS-07: Configuration Management

The system shall support tenant-specific configurations including schema extensions and retention policies.

10. VERIFICATION AND VALIDATION REQUIREMENTS (VV)

VV-01: Requirement Traceability Matrix

The project shall maintain a Requirements Traceability Matrix (RTM) mapping requirements to design modules and test cases.

VV-02: Unit Test Coverage

The system shall achieve at least 80% unit test coverage for critical services.

VV-03: Integration Testing

The system shall include integration tests for ingestion-to-publishing workflows.

VV-04: System Testing

The system shall include system tests validating end-to-end evidence-to-dataset generation.

VV-05: Security Testing

The system shall undergo penetration testing for SaaS deployments.

VV-06: Compliance Validation

The system shall support compliance validation audits through audit logs and traceability demonstrations.

VV-07: Dataset Reproducibility Test

The system shall demonstrate dataset regeneration capability producing identical outputs given identical evidence and configuration.

VV-08: Performance Testing

The system shall include load testing for ingestion throughput and OCR/extraction performance.

VV-09: Air-Gapped Validation

The system shall validate correct operation in offline deployment mode.

11. REQUIREMENT PRIORITIZATION (MoSCoW)

11.1 Must-Have Requirements

- FR-01 to FR-24
- FR-25 to FR-33
- FR-36 to FR-44
- SEC-01 to SEC-09
- DATA-01 to DATA-06
- NFR-01, NFR-02, NFR-06, NFR-12

11.2 Should-Have Requirements

- FR-35
- FR-45
- FR-50
- NFR-03, NFR-04, NFR-05
- OPS-04

11.3 Could-Have Requirements

- advanced workflow notifications
- automated rescan integration
- auto schema inference

12. CONCLUSION

This System Requirements Specification defines the complete set of functional, non-functional, security, integration, data, operational, and verification requirements for the Data Readiness System (DRS).

This document enables development teams to proceed with:

- Interface Control Documents (ICDs)
- Detailed design refinement
- Implementation
- Verification and Validation under the V-Model lifecycle